

①

$$\textcircled{a} N \times N = N^2 \text{ MAC}$$

$$\text{MAC} = 2 \text{ FLOPs}$$

$$\therefore 2N^2$$

⑧

$$N^2 \text{ FP32 at 4 bytes}$$

$$= 4N^2$$

⑨ s is the fraction of zeros then non zeros fraction is $(1-s)$

$$\therefore 2N^2(1-s)$$

$$\textcircled{d} 8N^2(1-s) + 4(N+1)$$

$$\textcircled{2} \frac{2N^2}{2N^2(1-s)} = \frac{1}{1-s}$$

$$\frac{1}{1-s} = 2 \quad \Rightarrow \quad 1-s = \frac{1}{2}$$

$s = \frac{1}{2}$ 50% of zeros needed for a 2x Flop speedup

③

$$4N^2 = 8N^2(1-s) + 4(N+1)$$
$$= \frac{4N^2}{4N^2} = \frac{8N^2(1-s) + 4N + 4}{4N^2}$$

$$= 1 - 2(1-s) + \frac{1}{N} + \frac{1}{N^2}$$

large N $\frac{1}{N}$ & $\frac{1}{N^2}$ are negligible

$$= 1 - 2(1-s) = \frac{1}{2} = 1 - s = \underline{s = \frac{1}{2}}$$

More than 50% zeros needed for CSR to use less memory than dense storage

④

$$4N^2 = 4(512)^2 = 1048576$$

$$8(512)^2(1-.9) + 4(512+1) = 211,767.2$$

$$\frac{1048576}{211,767.2} = 4.952$$